Transforms of Integrals and Integral Equations

A Self-Study Reference & Master Guide — Section 2.8

Contents

Part I — Theory & Worked Examples

Transforms of Integrals at a Glance: Core Sub-Topics

- **The Integral Operational Rule (2.8):** accumulating a function from 0 to t — that is, integrating in the time domain — corresponds to a single *division by* s in the transform domain. This is the exact mirror image of the derivative rule, where differentiation multiplied by s.
- **Volterra Integral Equations with $g(t) = 1$:** equations in which the unknown $x(t)$ appears *inside* an accumulation integral $\int_0^t x(\tau)\,d\tau$. Because the kernel is the constant 1, the integral is exactly the running integral of x, and the $F(s)/s$ rule turns the whole equation into one line of s-domain algebra.
- **Why division by s mirrors integration in t:** integration and differentiation are inverse operations in calculus, and the operators $\times s$ and $\div\, s$ are inverse operations in algebra. The Laplace transform is the dictionary that translates one inverse pair into the other.

Historical Origin: From Heaviside's Operators to Laplace's Rigor

These rules did not arrive as abstract mathematics — they were forged by engineers solving real circuits. In the late nineteenth century, the self-taught British physicist Oliver Heaviside developed his *operational calculus* to tame the telegraph equations governing long-distance signal transmission. His bold move was to treat the differential operator $p = \frac{d}{dt}$ as an ordinary *algebraic variable*: where calculus saw differentiation, Heaviside simply wrote "multiply by p," turning differential equations into algebra he could rearrange at will.

The decisive insight for this section is what happens to the *inverse* operation. If differentiation is multiplication by p, then its inverse — integration — must be division by p, i.e. the operator $\frac{1}{p}$. Heaviside used $\frac{1}{p}$ to mean "integrate" and obtained correct answers years before anyone could justify why. That justification came later, when the formal Laplace transform put his operators on rigorous footing: Heaviside's algebraic p became the transform variable s, and his operator $\frac{1}{p}$ became the rule $\frac{F(s)}{s}$ derived below. The division-by-s rule is thus a physicist's engineering shortcut, made mathematically exact.

2.8 The Integral Operational Rule

Throughout Section 2.7 we saw that differentiating a function in the time domain corresponds to *multiplying* its transform by s (and subtracting the initial value). Differentiation and integration are inverse operations, so we should expect integration to do the opposite: to *divide* the transform by s. That expectation is exactly correct, and it is one of the most useful labor-saving tools in the entire Laplace toolkit.

The Integral Operational Rule

If $\mathcal{L}\{f(t)\} = F(s)$, then the transform of the running integral of f is

$$\mathcal{L}\left\{ \int_0^t f(\tau)\, d\tau \right\} = \frac{F(s)}{s}.$$

Read in reverse, this is an inversion rule: dividing a known transform by s corresponds to integrating its inverse from 0 to t,

$$\mathcal{L}^{-1}\left\{ \frac{F(s)}{s} \right\} = \int_0^t f(\tau)\, d\tau, \qquad \text{where } f(\tau) = \mathcal{L}^{-1}\{F(s)\}\big|_{t=\tau}.$$

The variable of integration is written τ (a dummy variable) precisely so that it cannot be confused with the upper limit t. The upper limit t is the "real" time variable that survives into the answer; τ sweeps from 0 up to that t and is gone once the integral is evaluated.

A physical anchor — charge on a capacitor. Why should accumulation deserve its own operational rule? Consider an electrical capacitor. If $i(t)$ is the current flowing into it, the charge stored on its plates is not the current itself but the *total accumulated current up to the present moment* — the running integral

$$q(t) = \int_0^t i(\tau)\, d\tau.$$

Current is a *rate* (charge per second); charge is the *accumulation* of that rate over time. To find the charge in the Laplace domain, the Integral Operational Rule says you do not have to perform any integration at all: you simply transform the current and divide by s, $Q(s) = I(s)/s$. The same one move that converts current to charge converts velocity to position, or flow rate to total volume. Accumulation is everywhere in physics, and division by s handles all of it uniformly.

The n-fold Integral

The rule scales effortlessly. Integrating n times in a row — accumulating an accumulation, and so on — simply repeats the division by s once per integration:

$$\mathcal{L}\left\{\underbrace{\int\cdots\int}_{n} f(t)\,dt^{\,n}\right\} = \frac{F(s)}{s^{\,n}}.$$

Each nested integral contributes one factor of $\dfrac{1}{s}$, so n of them stack to $\dfrac{1}{s^{\,n}}$. This is exactly Heaviside's operator logic in miniature: the integration operator $\frac{1}{p}$ raised to the nth power is $\frac{1}{p^{\,n}}$. The single-integral rule $F(s)/s$ is the case $n = 1$; Problem 9 proves the case $n = 2$ explicitly.

The Proof — From the Derivative Rule, Step by Step

We will *derive* the integral rule rather than memorize it, building it entirely from the derivative rule we already trust. The idea: give the accumulation integral a name, differentiate it, and feed the result through the derivative rule.

Step 1 — Name the accumulation function. Define

$$g(t) = \int_0^t f(\tau)\,d\tau.$$

This $g(t)$ is the object whose transform we want; call that transform $G(s) = \mathcal{L}\{g(t)\}$. Our entire goal is to show $G(s) = F(s)/s$.

Step 2 — Differentiate g using the Fundamental Theorem of Calculus. The Fundamental Theorem of Calculus (Part 1) states that differentiating an integral with respect to its upper limit returns the integrand evaluated at that limit:

$$g'(t) = \frac{d}{dt}\int_0^t f(\tau)\,d\tau = f(t).$$

So g is an *antiderivative* of f — the specific one that starts at zero. This is the hinge of the whole proof.

Step 3 — Evaluate the initial value $g(0)$. Substitute $t = 0$ into the definition. The integral then runs from 0 to 0:

$$g(0) = \int_0^0 f(\tau)\,d\tau = 0.$$

An integral over an interval of zero width is zero, regardless of what f is. This vanishing initial value is what makes the rule so clean — there is no leftover boundary term.

Step 4 — Apply the Laplace derivative rule to g'. The first-derivative rule says $\mathcal{L}\{g'(t)\} = s\,G(s) - g(0)$. Substitute the two facts we just established — namely $g'(t) = f(t)$ from Step 2 and $g(0) = 0$ from Step 3:

$$\mathcal{L}\{g'(t)\} = s\,G(s) - g(0) \quad\Longrightarrow\quad \mathcal{L}\{f(t)\} = s\,G(s) - 0.$$

The left side is $\mathcal{L}\{f(t)\} = F(s)$ by definition, so

$$F(s) = s\,G(s).$$

Step 5 — Solve for $G(s)$. Divide both sides by s:

$$G(s) = \frac{F(s)}{s}.$$

Since $G(s) = \mathcal{L}\{\int_0^t f(\tau)\,d\tau\}$, this is exactly the Integral Operational Rule. ■

Worked Example 1: A Forward Transform via the Rule

Find $\mathcal{L}\left\{\int_0^t \cos(2\tau)\,d\tau\right\}$ in two ways and confirm they agree.

Method A — The integral rule. Here $f(\tau) = \cos(2\tau)$, whose transform is $F(s) = \dfrac{s}{s^2+4}$. The rule says divide by s:

$$\mathcal{L}\left\{\int_0^t \cos(2\tau)\,d\tau\right\} = \frac{F(s)}{s} = \frac{1}{s}\cdot\frac{s}{s^2+4} = \frac{1}{s^2+4}.$$

Method B — Integrate first, then transform. Evaluate the inner integral explicitly:

$$\int_0^t \cos(2\tau)\,d\tau = \left[\tfrac{1}{2}\sin(2\tau)\right]_0^t = \tfrac{1}{2}\sin(2t) - \tfrac{1}{2}\sin 0 = \tfrac{1}{2}\sin(2t).$$

Now transform the result, using $\mathcal{L}\{\sin(2t)\} = \dfrac{2}{s^2+4}$:

$$\mathcal{L}\left\{\tfrac{1}{2}\sin(2t)\right\} = \tfrac{1}{2}\cdot\frac{2}{s^2+4} = \frac{1}{s^2+4}.$$

The two methods agree exactly. Method A never required us to perform the integration — the division by s did the work.

2.9 Volterra Integral Equations with $g(t) = 1$

A **Volterra integral equation** is one in which the unknown function appears inside an integral with a variable upper limit t. When the kernel multiplying the unknown is simply 1, the integral $\int_0^t x(\tau)\,d\tau$ is precisely the running integral from the previous subsection, so the $F(s)/s$ rule applies directly to the unknown's own transform. This is the cleanest possible integral equation, and it is the perfect place to watch the rule in action.

Solving a $g(t) = 1$ Volterra Equation — The Recipe

Step 1 — Transform every term. Apply $\mathcal{L}$ to both sides. The accumulation integral $\int_0^t x(\tau)\,d\tau$ becomes $\dfrac{X(s)}{s}$ by the Integral Operational Rule.

Step 2 — Gather the $X(s)$ terms. Move every term containing $X(s)$ to one side and everything else to the other.

Step 3 — Factor out $X(s)$. The left side becomes $X(s)\cdot$(a rational expression in s).

Step 4 — Combine that factor over a common denominator, then multiply both sides by its reciprocal to isolate $X(s)$.

Step 5 — Decompose by partial fractions and invert term by term to recover $x(t)$.

Worked Example 2: A Volterra Integral Equation

Solve for $x(t)$:

$$x(t) - t = \int_0^t x(\tau)\, d\tau.$$

Step 1 — Transform every term. We take the Laplace transform of all three pieces. The transform of $x(t)$ is $X(s)$; the transform of t is $\frac{1}{s^2}$; and the accumulation integral on the right transforms by the Integral Operational Rule to $\frac{X(s)}{s}$:

$$X(s) - \frac{1}{s^2} = \frac{X(s)}{s}.$$

Step 2 — Gather the $X(s)$ terms on the left. Subtract $\frac{X(s)}{s}$ from both sides and add $\frac{1}{s^2}$ to both sides so that every X term sits on the left and the constant term sits on the right:

$$X(s) - \frac{X(s)}{s} = \frac{1}{s^2}.$$

Step 3 — Factor out $X(s)$. Both terms on the left share the common factor $X(s)$:

$$X(s)\left(1 - \frac{1}{s}\right) = \frac{1}{s^2}.$$

Step 4 — Combine the bracket over a common denominator. Write the 1 as $\frac{s}{s}$ so the two fractions inside the parentheses share the denominator s:

$$1 - \frac{1}{s} = \frac{s}{s} - \frac{1}{s} = \frac{s-1}{s}, \qquad \text{so} \qquad X(s)\left(\frac{s-1}{s}\right) = \frac{1}{s^2}.$$

Step 5 — Multiply by the reciprocal to isolate $X(s)$. The factor multiplying $X(s)$ is $\frac{s-1}{s}$; its reciprocal is $\frac{s}{s-1}$. Multiply both sides by that reciprocal:

$$X(s) = \frac{1}{s^2} \cdot \frac{s}{s-1}.$$

Now cancel one factor of s: the s in the numerator cancels one of the two factors of s in $s^2 = s \cdot s$, leaving a single s in the denominator:

$$X(s) = \frac{s}{s^2(s-1)} = \frac{1}{s(s-1)}.$$

The s-domain algebra is finished; the partial fraction work and inversion continue in the box below.

Worked Example 2 (continued): Partial Fractions and Inversion

We carry $X(s) = \dfrac{1}{s(s-1)}$ forward, decompose it, and invert.

Step 6 — Set up the partial fraction decomposition. The denominator $s(s-1)$ has two distinct linear factors, so we write

$$\frac{1}{s(s-1)} = \frac{A}{s} + \frac{B}{s-1}.$$

Multiply both sides by the common denominator $s(s-1)$ to clear all fractions:

$$1 = A(s-1) + B\,s.$$

Step 7 — Solve for A and B. This identity must hold for *every* value of s, so we substitute strategic values that kill one unknown at a time.

Let $s = 0$: the $B\,s$ term vanishes, leaving

$$1 = A(0-1) + B(0) = -A \quad \Longrightarrow \quad A = -1.$$

Let $s = 1$: the $A(s-1)$ term vanishes, leaving

$$1 = A(1-1) + B(1) = B \quad \Longrightarrow \quad B = 1.$$

So the decomposition is

$$X(s) = \frac{1}{s(s-1)} = \frac{-1}{s} + \frac{1}{s-1}.$$

Step 8 — Invert term by term. Using $\mathcal{L}^{-1}\{1/s\} = 1$ and $\mathcal{L}^{-1}\{1/(s-1)\} = e^t$:

$$x(t) = -1 + e^t = e^t - 1.$$

Step 9 — Verify in the original equation. A quick check confirms the answer. With $x(t) = e^t - 1$, the left side is $x(t) - t = e^t - 1 - t$. The right side is

$$\int_0^t (e^\tau - 1)\,d\tau = \left[e^\tau - \tau\right]_0^t = (e^t - t) - (e^0 - 0) = e^t - t - 1,$$

which matches the left side exactly. ✓

Divide the Whole Transform by s, Not Just One Piece

When you apply the $F(s)/s$ rule, the *entire* transform $F(s)$ goes over s — not just a single factor of it. For example,

$$\mathcal{L}\left\{\int_0^t (\tau + e^\tau)\, d\tau\right\} = \frac{1}{s}\left(\frac{1}{s^2} + \frac{1}{s-1}\right) = \frac{1}{s^3} + \frac{1}{s(s-1)},$$

not $\frac{1}{s^3} + \frac{1}{s-1}$. The division distributes across the whole sum because $F(s)$ is the transform of the whole integrand.

Key Takeaway

Integration in t becomes division by s. The running integral $\int_0^t f(\tau)\, d\tau$ transforms to $\dfrac{F(s)}{s}$ — the exact inverse of the derivative rule, in which differentiation multiplied by s. Any equation that buries its unknown inside an accumulation integral is therefore just an algebra problem in disguise: divide by s, isolate $X(s)$, and invert.

Part II — Practice Set

Work each problem before turning to the complete solutions in Part III. Problems 1–3 drill the forward rule, 4–5 the inverse rule, 6–8 build to full integral and integro-differential equations, and 9–10 ask you to prove the rule two different ways.

Problem 1: Forward Transform of an Integral.
Find $\mathcal{L}\left\{\int_0^t \sin(3\tau)\,d\tau\right\}$.

Problem 2: Forward Transform of an Integral.
Find $\mathcal{L}\left\{\int_0^t e^{-2\tau}\,d\tau\right\}$.

Problem 3: Forward Transform of an Integral.
Find $\mathcal{L}\left\{\int_0^t \tau^2\,d\tau\right\}$ using the integral rule, and confirm by integrating first.

Problem 4: Inverse Transform via the $1/s$ Rule.
Find $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+4)}\right\}$.

Problem 5: Inverse Transform via the $1/s$ Rule.
Find $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s+1)}\right\}$.

Problem 6: A Volterra Integral Equation.
Solve for $x(t)$:

$$x(t) = 1 + \int_0^t x(\tau)\,d\tau.$$

Problem 7: A Volterra Integral Equation.
Solve for $x(t)$:

$$x(t) = t^2 - \int_0^t x(\tau)\,d\tau.$$

Problem 8: An Integro-Differential Equation.
Solve for $x(t)$, given $x(0) = 2$:

$$x'(t) + \int_0^t x(\tau)\,d\tau = 1.$$

Problem 9 (Proof): The Double-Integral Rule.
Prove that

$$\mathcal{L}\left\{\int_0^t \int_0^\tau f(u)\,du\,d\tau\right\} = \frac{F(s)}{s^2}.$$

Problem 10 (Proof): The Rule via the Convolution Theorem.
Prove the Integral Operational Rule by recognizing that $\int_0^t f(\tau)\,d\tau$ is the convolution of $f(t)$ with the constant function $g(t) = 1$.

Part III — Complete Solutions

Solution 1: $\mathcal{L}\left\{\int_0^t \sin(3\tau)\,d\tau\right\}$**.**

The integrand is $f(\tau) = \sin(3\tau)$, with transform $F(s) = \dfrac{3}{s^2+9}$. The integral rule divides this by s:

$$\mathcal{L}\left\{\int_0^t \sin(3\tau)\,d\tau\right\} = \frac{F(s)}{s} = \frac{1}{s}\cdot\frac{3}{s^2+9} = \frac{3}{s(s^2+9)}.$$

Cross-check. Integrating first gives $\int_0^t \sin(3\tau)\,d\tau = \left[-\frac{1}{3}\cos(3\tau)\right]_0^t = \frac{1}{3}(1-\cos 3t)$, whose transform is $\frac{1}{3}\left(\frac{1}{s} - \frac{s}{s^2+9}\right) = \frac{1}{3}\cdot\frac{(s^2+9)-s^2}{s(s^2+9)} = \dfrac{3}{s(s^2+9)}$, in agreement.

Solution 2: $\mathcal{L}\left\{\int_0^t e^{-2\tau}\,d\tau\right\}$**.**

The integrand is $f(\tau) = e^{-2\tau}$, with transform $F(s) = \dfrac{1}{s+2}$. Divide by s:

$$\mathcal{L}\left\{\int_0^t e^{-2\tau}\,d\tau\right\} = \frac{1}{s}\cdot\frac{1}{s+2} = \frac{1}{s(s+2)}.$$

Cross-check. The inner integral is $\int_0^t e^{-2\tau}\,d\tau = \left[-\frac{1}{2}e^{-2\tau}\right]_0^t = \frac{1}{2}(1-e^{-2t})$, whose transform is $\frac{1}{2}\left(\frac{1}{s} - \frac{1}{s+2}\right) = \frac{1}{2}\cdot\frac{(s+2)-s}{s(s+2)} = \dfrac{1}{s(s+2)}$. ✓

Solution 3: $\mathcal{L}\left\{\int_0^t \tau^2\,d\tau\right\}$**.**

Via the rule. The integrand is $f(\tau) = \tau^2$, with transform $F(s) = \dfrac{2!}{s^3} = \dfrac{2}{s^3}$. Divide by s:

$$\mathcal{L}\left\{\int_0^t \tau^2\,d\tau\right\} = \frac{1}{s}\cdot\frac{2}{s^3} = \frac{2}{s^4}.$$

Integrating first. $\int_0^t \tau^2\,d\tau = \left[\frac{1}{3}\tau^3\right]_0^t = \frac{t^3}{3}$, and $\mathcal{L}\{\frac{t^3}{3}\} = \frac{1}{3}\cdot\dfrac{3!}{s^4} = \frac{1}{3}\cdot\dfrac{6}{s^4} = \dfrac{2}{s^4}$. The two routes agree.

Solution 4: $\mathcal{L}^{-1}\left\{\dfrac{1}{s(s^2+4)}\right\}$**.**

Step 1 — Recognize the $1/s$ structure. The factor $\dfrac{1}{s}$ out front signals integration of the inverse of the remaining factor. Write

$$\frac{1}{s(s^2+4)} = \frac{1}{s}\cdot\frac{1}{s^2+4} = \frac{1}{s}\,F(s), \qquad F(s) = \frac{1}{s^2+4}.$$

Step 2 — Invert the inner transform. Since $\dfrac{2}{s^2+4} \to \sin(2t)$, we have $\dfrac{1}{s^2+4} \to \frac{1}{2}\sin(2t)$, so $f(\tau) = \frac{1}{2}\sin(2\tau)$.

Step 3 — Integrate from 0 to t. By the inverse integral rule, $\mathcal{L}^{-1}\{F(s)/s\} = \int_0^t f(\tau)\,d\tau$:

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+4)}\right\} = \int_0^t \tfrac{1}{2}\sin(2\tau)\,d\tau = \tfrac{1}{2}\left[-\tfrac{1}{2}\cos(2\tau)\right]_0^t = \tfrac{1}{4}\big(1-\cos 2t\big).$$

Cross-check by partial fractions. We rebuild the same answer from scratch using a partial fraction decomposition, showing every algebraic step.

(a) Set up the form. The denominator has a linear factor s and an irreducible quadratic factor s^2+4. A linear factor gets a constant numerator; an irreducible quadratic gets a *linear* numerator $Bs+C$:

$$\frac{1}{s(s^2+4)} = \frac{A}{s} + \frac{Bs+C}{s^2+4}.$$

(b) Clear the denominators. Multiply both sides by the full denominator $s(s^2+4)$. On the right, the first term keeps s^2+4 and the second keeps s:

$$1 = A(s^2+4) + (Bs+C)\,s.$$

(c) Expand and group by powers of s. Distribute every product, then collect like powers:

$$1 = As^2 + 4A + Bs^2 + Cs = (A+B)\,s^2 + C\,s + 4A.$$

(d) Equate coefficients. The left side is the constant 1, i.e. $0\cdot s^2 + 0\cdot s + 1$. Matching the coefficient of each power of s on both sides gives one equation per power:

$$\underbrace{A+B=0}_{s^2\text{ terms}},\qquad \underbrace{C=0}_{s^1\text{ terms}},\qquad \underbrace{4A=1}_{s^0\text{ terms}}.$$

(e) Solve the system. The constant equation gives A immediately; back-substitute into the s^2 equation for B:

$$4A=1 \;\Rightarrow\; A=\tfrac{1}{4},\qquad A+B=0 \;\Rightarrow\; B=-A=-\tfrac{1}{4},\qquad C=0.$$

So the decomposition is

$$\frac{1}{s(s^2+4)} = \frac{1}{4}\cdot\frac{1}{s} - \frac{1}{4}\cdot\frac{s}{s^2+4}.$$

(f) Invert term by term. Using $\mathcal{L}^{-1}\{1/s\}=1$ and $\mathcal{L}^{-1}\{s/(s^2+4)\}=\cos 2t$:

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+4)}\right\} = \frac{1}{4} - \frac{1}{4}\cos 2t = \tfrac{1}{4}\big(1-\cos 2t\big),$$

exactly matching the result from the $1/s$ rule above.

Solution 5: $\mathcal{L}^{-1}\left\{\dfrac{1}{s^2(s+1)}\right\}$.

Step 1 — Peel off one factor of $1/s$. Write

$$\frac{1}{s^2(s+1)} = \frac{1}{s}\cdot\frac{1}{s(s+1)} = \frac{1}{s}\,F(s),\qquad F(s)=\frac{1}{s(s+1)}.$$

Step 2 — Invert the inner transform $F(s)$. Decompose $\dfrac{1}{s(s+1)} = \dfrac{A}{s} + \dfrac{B}{s+1}$. Clearing gives $1 = A(s+1) + Bs$. At $s = 0$: $A = 1$. At $s = -1$: $1 = -B \Rightarrow B = -1$. Hence

$$F(s) = \frac{1}{s} - \frac{1}{s+1} \quad \Longrightarrow \quad f(\tau) = 1 - e^{-\tau}.$$

Step 3 — Integrate from 0 to t.

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2(s+1)}\right\} = \int_0^t \left(1 - e^{-\tau}\right) d\tau = \left[\tau + e^{-\tau}\right]_0^t = \left(t + e^{-t}\right) - \left(0 + 1\right) = t - 1 + e^{-t}.$$

Cross-check by full partial fractions. We decompose the whole expression at once, showing every step.

(a) Set up the form. The factor s is *repeated* (it appears as s^2), so a repeated linear factor requires *one term for each power* up to its multiplicity — both $\dfrac{A}{s}$ and $\dfrac{B}{s^2}$ — plus a term for the distinct factor $s + 1$:

$$\frac{1}{s^2(s+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1}.$$

(b) Clear the denominators. Multiply both sides by $s^2(s+1)$. The $\dfrac{A}{s}$ term keeps $s(s+1)$, the $\dfrac{B}{s^2}$ term keeps $(s+1)$, and the $\dfrac{C}{s+1}$ term keeps s^2:

$$1 = A\, s(s+1) + B(s+1) + C\, s^2.$$

(c) Use strategic roots for B and C. Substituting the roots of the denominator zeroes out two of the three unknowns at a time.

Let $s = 0$: the A and C terms carry a factor of s and vanish, leaving

$$1 = B(0+1) = B \quad \Longrightarrow \quad B = 1.$$

Let $s = -1$: the A and B terms carry a factor of $(s+1)$ and vanish, leaving

$$1 = C(-1)^2 = C \quad \Longrightarrow \quad C = 1.$$

(d) Find A by matching the leading coefficient. The substitution trick cannot reach A directly (no remaining convenient root), so expand and compare the highest power. Expanding the right side,

$$1 = A(s^2 + s) + Bs + B + Cs^2 = (A + C)\, s^2 + (A + B)\, s + B.$$

The left side has no s^2 term, so the coefficient of s^2 must be zero:

$$A + C = 0 \quad \Longrightarrow \quad A = -C = -1.$$

(e) Assemble and invert. With $A = -1$, $B = 1$, $C = 1$:

$$\frac{1}{s^2(s+1)} = -\frac{1}{s} + \frac{1}{s^2} + \frac{1}{s+1}.$$

Inverting term by term with $\mathcal{L}^{-1}\{1/s\} = 1$, $\mathcal{L}^{-1}\{1/s^2\} = t$, and $\mathcal{L}^{-1}\{1/(s+1)\} = e^{-t}$:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2(s+1)}\right\} = -1 + t + e^{-t} = t - 1 + e^{-t},$$

matching the result from the $1/s$ rule.

Solution 6: $x(t) = 1 + \displaystyle\int_0^t x(\tau)\,d\tau$**.**

Step 1 — Transform. With $\mathcal{L}\{1\} = \frac{1}{s}$ and the integral rule turning $\int_0^t x(\tau)\,d\tau$ into $\dfrac{X(s)}{s}$:

$$X(s) = \frac{1}{s} + \frac{X(s)}{s}.$$

Step 2 — Gather X terms and factor.

$$X(s) - \frac{X(s)}{s} = \frac{1}{s} \quad \Longrightarrow \quad X(s)\left(1 - \frac{1}{s}\right) = \frac{1}{s}.$$

Step 3 — Common denominator in the bracket.

$$1 - \frac{1}{s} = \frac{s-1}{s}, \qquad \text{so} \qquad X(s)\left(\frac{s-1}{s}\right) = \frac{1}{s}.$$

Step 4 — Multiply by the reciprocal.

$$X(s) = \frac{1}{s} \cdot \frac{s}{s-1} = \frac{1}{s-1}.$$

Step 5 — Invert.

$$x(t) = e^t.$$

Check. $1 + \int_0^t e^\tau\,d\tau = 1 + (e^t - 1) = e^t = x(t)$. ✓

Solution 7: $x(t) = t^2 - \displaystyle\int_0^t x(\tau)\,d\tau$**.**

Step 1 — Transform. With $\mathcal{L}\{t^2\} = \dfrac{2}{s^3}$ and the integral $\to \dfrac{X(s)}{s}$:

$$X(s) = \frac{2}{s^3} - \frac{X(s)}{s}.$$

Step 2 — Gather X terms and factor. Add $\dfrac{X(s)}{s}$ to both sides:

$$X(s) + \frac{X(s)}{s} = \frac{2}{s^3} \quad \Longrightarrow \quad X(s)\left(1 + \frac{1}{s}\right) = \frac{2}{s^3}.$$

Step 3 — Common denominator in the bracket.

$$1 + \frac{1}{s} = \frac{s+1}{s}, \qquad \text{so} \qquad X(s)\left(\frac{s+1}{s}\right) = \frac{2}{s^3}.$$

Step 4 — Multiply by the reciprocal and cancel one s.

$$X(s) = \frac{2}{s^3} \cdot \frac{s}{s+1} = \frac{2}{s^2(s+1)}.$$

Step 5 — Partial fractions. Set $\dfrac{2}{s^2(s+1)} = \dfrac{A}{s} + \dfrac{B}{s^2} + \dfrac{C}{s+1}$. Clear denominators:

$$2 = A\,s(s+1) + B(s+1) + C\,s^2.$$

At $s = 0$: $2 = B(1) \Rightarrow B = 2$. At $s = -1$: $2 = C(-1)^2 = C \Rightarrow C = 2$. Matching s^2 coefficients: $0 = A + C \Rightarrow A = -2$. So

$$X(s) = \frac{-2}{s} + \frac{2}{s^2} + \frac{2}{s+1}.$$

Step 6 — Invert term by term.

$$x(t) = -2 + 2t + 2e^{-t}.$$

Check. $\int_0^t (-2 + 2\tau + 2e^{-\tau})\,d\tau = \big[-2\tau + \tau^2 - 2e^{-\tau}\big]_0^t = -2t + t^2 - 2e^{-t} + 2$. Then $t^2 - (\text{this}) = t^2 + 2t - t^2 + 2e^{-t} - 2 = 2t - 2 + 2e^{-t} = x(t)$. ✓

Solution 8: $x'(t) + \displaystyle\int_0^t x(\tau)\,d\tau = 1, \quad x(0) = 2$.

Step 1 — Transform each term, keeping the initial value. The derivative rule is $\mathcal{L}\{x'\} = sX(s) - x(0)$. With $x(0) = 2$ this becomes $sX(s) - 2$ — the initial value does *not* drop out this time. The integral rule gives $\int_0^t x \to \dfrac{X(s)}{s}$, and $\mathcal{L}\{1\} = \dfrac{1}{s}$:

$$sX(s) - 2 + \frac{X(s)}{s} = \frac{1}{s}.$$

Step 2 — Move the constant -2 to the right side. Add 2 to both sides so that only $X(s)$ terms remain on the left:

$$sX(s) + \frac{X(s)}{s} = \frac{1}{s} + 2.$$

Step 3 — Factor out $X(s)$ on the left.

$$X(s)\left(s+\frac{1}{s}\right)=\frac{1}{s}+2.$$

Step 4 — Common denominator on *both* sides. The bracket on the left combines over s, and the right side combines over s as well by writing $2=\dfrac{2s}{s}$:

$$s+\frac{1}{s}=\frac{s^2+1}{s},\qquad \frac{1}{s}+2=\frac{1}{s}+\frac{2s}{s}=\frac{2s+1}{s},$$

so the equation becomes

$$X(s)\left(\frac{s^2+1}{s}\right)=\frac{2s+1}{s}.$$

Step 5 — Multiply by the reciprocal and cancel. Multiply both sides by $\dfrac{s}{s^2+1}$; the factors of s cancel cleanly:

$$X(s)=\frac{2s+1}{s}\cdot\frac{s}{s^2+1}=\frac{2s+1}{s^2+1}.$$

Step 6 — Split into table-ready pieces. Separate the numerator over the common denominator:

$$X(s)=\frac{2s+1}{s^2+1}=2\cdot\frac{s}{s^2+1}+\frac{1}{s^2+1}.$$

Step 7 — Invert. Using $\mathcal{L}^{-1}\{s/(s^2+1)\}=\cos t$ and $\mathcal{L}^{-1}\{1/(s^2+1)\}=\sin t$:

$$x(t)=2\cos t+\sin t.$$

Check. The initial condition holds: $x(0)=2\cos 0+\sin 0=2$. ✓Differentiating, $x'(t)=-2\sin t+\cos t$, and $\int_0^t(2\cos\tau+\sin\tau)\,d\tau=\big[2\sin\tau-\cos\tau\big]_0^t=2\sin t-\cos t+1$. Adding, $x'(t)+\int_0^t x\,d\tau=(-2\sin t+\cos t)+(2\sin t-\cos t+1)=1$. ✓

Solution 9 (Proof): The Double-Integral Rule.

We want $\mathcal{L}\left\{\displaystyle\int_0^t\int_0^\tau f(u)\,du\,d\tau\right\}=\dfrac{F(s)}{s^2}$.

Step 1 — Name the inner integral. Let

$$h(\tau)=\int_0^\tau f(u)\,du.$$

By the single-integral rule proved in Part I, $\mathcal{L}\{h(\tau)\}=H(s)=\dfrac{F(s)}{s}$.

Step 2 — The outer integral is the running integral of h. The full expression is

$$\int_0^t h(\tau)\,d\tau.$$

This is exactly the running integral of h, so the single-integral rule applies a *second* time, now to h:

$$\mathcal{L}\left\{\int_0^t h(\tau)\,d\tau\right\} = \frac{H(s)}{s}.$$

Step 3 — Substitute $H(s) = F(s)/s$.

$$\frac{H(s)}{s} = \frac{1}{s}\cdot\frac{F(s)}{s} = \frac{F(s)}{s^2}.$$

Each integration contributed one factor of $\frac{1}{s}$; two integrations give $\frac{1}{s^2}$. By induction, n nested integrations give $\frac{F(s)}{s^n}$. ■

Solution 10 (Proof): The Rule via the Convolution Theorem.
Step 1 — Recall the convolution theorem. For suitable f and g,

$$\mathcal{L}\{(f*g)(t)\} = F(s)\,G(s), \qquad (f*g)(t) = \int_0^t f(\tau)\,g(t-\tau)\,d\tau.$$

Step 2 — Choose $g(t) = 1$. The constant function $g(t) \equiv 1$ has $g(t-\tau) = 1$ for every τ. Substituting into the convolution integral:

$$(f*1)(t) = \int_0^t f(\tau)\cdot 1\,d\tau = \int_0^t f(\tau)\,d\tau.$$

So the running integral of f is *literally* the convolution of f with the constant 1.

Step 3 — Apply the convolution theorem. The transform of $g(t) = 1$ is $G(s) = \frac{1}{s}$. Therefore

$$\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \mathcal{L}\{(f*1)(t)\} = F(s)\,G(s) = F(s)\cdot\frac{1}{s} = \frac{F(s)}{s}.$$

This reproduces the Integral Operational Rule, now seen as the special case of convolution against the constant 1. ■

Bridge to the Next Unit

That "constant function $g(t) = 1$" is not as innocent as it looks. A function that equals 1 for all $t \geq 0$ (and 0 before) is formally the Heaviside step function, written $u(t)$ or $u_0(t)$ — the very function Heaviside introduced to switch a signal on at $t = 0$. Step 2 above therefore carries a deeper reading: the running integral $\int_0^t f(\tau)\,d\tau$ is precisely $f * u_0$, the convolution of f with the unit step. In one sentence:

Integration is simply convolution with the unit step function $u_0(t)$.

When the next unit takes up the step function and piecewise forcing in earnest, this identity is the hinge that connects it back to everything proved here.

Closing Takeaway

Two independent proofs — one through the derivative rule and the Fundamental Theorem of Calculus, the other through the convolution theorem with $g(t) = 1$ — converge on the same identity $\mathcal{L}\{\int_0^t f\,d\tau\} = F(s)/s$. Whenever an unknown hides inside an accumulation integral, divide by s, isolate $X(s)$, and invert: integration in t is division by s.